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Isovalent Mesh Networking

One Mesh to Connect Them All

Businesses today demand seamless connectivity across Kubernetes, virtual machines, and physical servers, all while navigating hybrid and multi-cloud environments. Isovalent Mesh Networking provides a universal networking layer that simplifies operations, enhances security, and ensures consistent application performance.

By eliminating the need for fragmented network stacks, the Isovalent Enterprise Platform reduces operational complexity, strengthens reliability, and enables enterprises to modernize their infrastructure without sacrificing performance or security.

The Future of Connectivity

Traditional networks struggle to meet the demands of hybrid and multi-cloud environments. Fragmentation across platforms leads to inefficiencies, increased vulnerabilities, and slower business growth. These challenges make it difficult for organizations to maintain operational consistency and scale effectively.

Isovalent Mesh Networking addresses these issues by consolidating cloud-native and legacy systems into a single, scalable solution. Isovalent simplifies connectivity by creating a unified network fabric, streamlines security with zero-trust principles and encryption, and improves performance through real-time observability and automated troubleshooting. The result is faster deployments, stronger protection against risks, and more reliable application performance.

Simplify Network Management

Unify Kubernetes, VMs, and physical servers into a single network layer to reduce complexity and accelerate operations.

Strengthen Infrastructure Security

Protect workloads with zero-trust principles, micro-segmentation, and transparent encryption for secure, scalable operations.

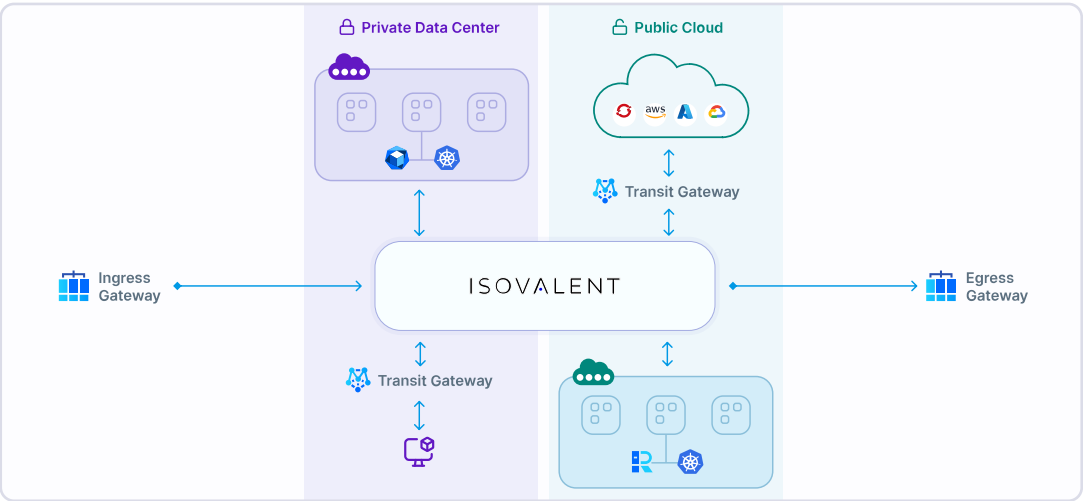
Extract End-to-End Observability

Extract real-time insights into application performance and infrastructure health with deep monitoring and automated diagnostics.

Achieve Multi-Cloud Consistency

Implement consistent operations across unique clouds, data centers, and edge environments with unified networking, security, and observability.

> *Before [Isovalent], our teams had to manage different networking configurations per cloud. Now, networking is standardized, and our developers don't have to think about cloud-specific differences.*
>
> **- Alvaro Aleman, Staff Software Engineer at Confluent**
>
> CONFLUENT



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Consistent Outcomes Across Any Infrastructure

Isovalent chosen as the standard network interface by all major cloud providers, making it an industry standard.

2020

Google announces Isovalent + eBPF as the new networking data plane for GKE

2021

AWS picks Isovalent for Networking + Security on EKS anywhere

2022

Announcing Azure CNI Powered by Isovalent

Seamlessly Connect Workloads Across Any Cloud

Extending cloud-native networking to traditional, hybrid, and multi-cloud environments.

Static Egress Gateway: Control and secure outbound traffic with centralized policy enforcement.

On-Prem Integration: Extend Kubernetes networking to traditional and hybrid environments.

Transit Gateway: Deliver consistent connectivity across cloud-native and legacy systems.

VMware Bridge: Integrate Kubernetes networking with VMware environments.

Safeguard Applications with Identity-Based Policies

Advanced Network Policy: Define and enforce granular network access rules across environments.

Transparent Encryption: Secure data in transit with IPsec and WireGuard encryption. (FIPS compliant).

Multi-Tenant Network Isolation: Secure multi-tenant environments with identity-aware, per-tenant segmentation.

Microsegmentation: Enforce fine-grained workload isolation to minimize attack surfaces.

Accelerate Application Delivery & Load Balancing

Ensure low-latency, high-availability application delivery across cloud, hybrid, and on-prem environments.

Scalable Load Balancing (and Ingress): Optimize traffic distribution with high-performance, topology-aware load balancing.

Bandwidth and Latency Optimization: Improve network efficiency and reduce latency for mission-critical applications.

Multi-Environment Load Balancing: Ensure consistent load balancing across Kubernetes, public cloud, VMs, and bare metal.

Customizable Policies: Allows for the creation and implementation of custom policies to meet specific security needs.

Frictionless Developer Experience: Automated deployment and API-based configuration, supporting modern DevOps practices.

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Why Choose the Isovalent Enterprise Platform Over Cilium OSS?

Isovalent Enterprise is a hardened, production-grade distribution of the open source Cilium project, designed to meet the demands of mission-critical enterprise environments. It delivers:

Extended Lifecycle Support: Stable, tested releases with long-term support, hotfixes, and backports for critical issues.

24/7 Enterprise-Grade Support: Direct access to the creators of Cilium and eBPF for expert guidance and rapid issue resolution.

Advanced Enterprise Features:

- High Availability & Scalability
- Security & Compliance at Scale
- Proactive support (TAM).

With the Isovalent Enterprise Platform, organizations gain the performance and innovation of open source Cilium, plus the security, stability, and support required to operate confidently at scale. It's not just open source ready for production. It's open source made enterprise-grade.

> *Isovalent is the go-to expert on eBPF. Their expertise and understanding of how to use these technologies have been crucial in helping us unpack the data eBPF unlocks.*
>
> **- Blake Romero, Senior Software Engineer, Imagine Learning**

imagine learning

Lightweight eBPF-based architecture

Ease of deployment, on any application

Dynamic control for security and platform teams

Take the next step with Isovalent Mesh Networking!

isovalent.site/discover-lab: Get started with Isovalent labs in seconds. Explore use cases around Network Observability, Network Policies, Transparent Encryption, Mutual Authentication, or Runtime Security and Enforcement.

isovalent.site/demo: Book your personalized session today and dive into the full potential of the Isovalent Enterprise Platform.

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Isovalent is the company founded by the creators of Cilium and eBPF. We build open source software and enterprise solutions solving networking, security, and observability needs for modern cloud native infrastructure. Google (GKE, Anthos), Amazon (EKS-A), and Microsoft (AKS) have all adopted Cilium to provide networking and security for Kubernetes. Cilium is used by platform engineering teams such as Adobe, Bell Canada, ByteDance, Capital One, Datadog, IKEA, Schuberg Philis, and Sky.

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